Textures Unlimited Recolour Depot

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By Manwith Noname, April 22, 2018 in Add-on Development

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Manwith Noname
Rocketry Enthusiast

Members
219
320 posts

Posted April 22, 2018 (edited)

Report post

Textures Unlimited Recolour Depot

Still thinking in progress...comic sans for comic...hmm, maybe not.

What is this?

> This is a continuation / adaptation / expansion of previous work I undertook for the mod KerbPaint many moons ago. It now uses shaders made available via Textures Unlimited to bring user control over the colour of many stock parts.

Is it finished?

Sadly no, still thinking but you can check out the current state below.

Can we see it?

The following pics show the colour mapping for stock parts.

Reveal hidden contents

Here's something a bit more...tasteful.....maybe.

Reveal hidden contents

Known issues:

Stock part variants are a pain in the bum.

Installation and Dependencies:

Ensure both Textures Unlimited and Module Manager are installed.

Download the Stock Recolour pack below and place it appropriately within your game directory.

Extra packs can then be installed if you have the mods they cover. Correct installation will result in all packs being placed within a dedicated Textures Unlimited Recolour Depot folder so your GameData isn't taken over by them.

NOTE: The config files specify paths to textures. If you alter the path of a pack or it's parent mod, the universe will collapse.

Downloads

Stock Recolour

This is the core pack. It applies the TU texture switch module and enables switching between default stock shaders and the PBR recolour shader of TU for stock parts.

Stock Extras

This pack adds a metallic set to the texture switch module. You know, the shiny shiny all covering thing that gets people gooey. Needs the Stock Recolour pack.

MK2 Expansion

This pack applies the stock colour maps to parts in the MK2 Expansion mod by SuicidalInsanity. The Stock Recolour pack above is required. MK2 Expansion mod thread here.

MK3 Expansion

This pack applies the stock colour maps to parts in the MK3 Expansion mod by SuicidalInsanity. The Stock Recolour pack above is required. MK3 Expansion mod thread here.

OPT Spaceplane

This pack applies the TU texture Switch module to OPT Spaceplane parts to enable switching between the stock and recolour shaders. Currently requires the core Stock Recolour pack, this may change. OPT Spaceplane mod thread here.

Licensing

Unless otherwise stated.

Credits:

@Texel If they had not created KerbPaint, I would never have made masks for Porkjets MK2 parts (yes, before they were stock).

@Shadowmage for Textures Unlimited (and other great stuff too) without which this wouldn't exist.

@Electrocutor for endless tinkering and posting in the forum which helped me understand TU better.

@SuicidalInsanity for stock themed part expansions and actually implementing some of my suggestions in their

mods. Didn't expect that.

@K.Yeon for OPT. Love the design of this pack.

@SQUAD I guess...dunno why...let me think about that.

Other peeps no doubt.

Edited September 26, 2018 by Manwith Noname

Quote

ISE, Drew Kerman, Colonel Cbplayer and 19 others like this

Money doesn't make the world go round...

...gravity does.

bcink

JWST Kerbal Engineer

Members

1,377

438 posts

Posted April 23, 2018

Report post

Very promising. Are you re-making all stock texture files to base colors for T.U. adaptation?

Quote

James Webb [WIP] Thread Kerbalow Aerospace [WIP] Thread DSSHU [WIP] Thread Making Alternate History [WIP] Thread Duna Direct [WIP] Thread

RedParadize

Spacecraft Engineer

Posted April 23, 2018

Report post

That's awesome!

Members
244
775 posts

Quote

Manwith Noname likes this

Manwith Noname
Rocketry Enthusiast

Members
219
320 posts

Posted April 24, 2018 (edited)

Report post

> **On 4/23/2018 at 2:11 AM, bcink said:**
>
> Very promising. Are you re-making all stock texture files to base colors for T.U. adaptation?

Essentially, yes, though that is a part of the "thinking in progress" title. I was going to add a roadmap to the opening post at some point when I've though it out some more but off the top of my head it looks something like this...

Phase 1 - Get maps made / tidy for parts and working in game.

Phase 2 - Create dedicated specular / metallic maps.

Phase 3 - Normals?

There's also something I need to decide before starting phase 2 in earnest. Textures for masked shading without "tinting mode" appear to all be scaled around 127 / 128 RGB values while the tinting mode expects the max value to be 255...though I might be mistaken and misread things.

After converting configs to work in 1.4 I started dabbling in Phase 2 but I will probably revert to how things were working prior for any test release as converting every parts textures is quite a task to do and while without them there are issues, having a mismatch of how parts and textures are setup is probably more annoying. I should perhaps also mention, while getting stock sorted should probably be the priority, I have some mods covered too and would like to cover more.

Reveal hidden contents

Edited April 24, 2018 by Manwith Noname

Quote ISE and bcink like this

Money doesn't make the world go round...

...gravity does.

bcink

JWST Kerbal Engineer

Members

1,377

438 posts

Posted April 24, 2018

Report post

Sweet!

Quote Manwith Noname likes this

James Webb [WIP] Thread Kerbalow Aerospace [WIP] Thread DSSHU [WIP] Thread Making Alternate History [WIP] Thread Duna Direct [WIP] Thread

Manwith Noname

Rocketry Enthusiast

Members

219

320 posts

Posted April 26, 2018

Report post

@bcink I have a question actually. How much work would be involved in copying a collider from one model to another? More precisely in this instance, the new Rockomax fuel tanks. The only way it seems user defined recolouring is going to work on them and stick in the flight scene is by removing part variants (yes, I realise that might upset people) but in doing so it became evident the collider must be copied over by the game somehow as it is not present when the orange tank is setup as a single part.

More generally, I'm curious if people look at their craft

closely enough that differences like this would benefit them?

Reveal hidden contents

I really need to stop feature creeping.

Quote

bcink likes this

Money doesn't make the world go round...

...gravity does.

Shadowmage

Sr. Spacecraft Engineer

Members

6,304

4,341 posts

Posted April 26, 2018

Report post

> **On 4/26/2018 at 2:46 PM, Manwith Noname said:**
>
> The only way it seems user defined recolouring is going to work on them and stick in the flight scene is by removing part variants (yes, I realise that might upset people) but in doing so it became evident the collider must be copied over by the game somehow as it is not present when the orange tank is setup as a single part.

Not going to be doable without custom plugin code. Stock code doesn't support that level of model manipulation as far as I'm aware. You -might- be able to pull it off with one of the PartSwitch modules/mods (e.g. B9PartSwitch), but this would highly depend on exactly how the stock models are setup, and how the part-switch module handles sub-meshes.

After I'm finished updating SSTU for KSP 1.4+ I intend on digging into a couple of the larger outstanding issues with TU -- DirectX reflections, and the new stock part-variant mess (yes, it is a huge mess). So I would say hold off on the part-variant stuff for a bit longer, as I'll hopefully get some sort of texture switch (or just a recoloring) module developed for them in the (near'ish) future.

Quote

RedParadize, Manwith Noname, bcink and 1 other like this

SS
Low Part Co
Lifters, Orbiters,
Modular Parts an
(Deve

bcink

JWST Kerbal Engineer

Members
1,377
438 posts

Posted April 26, 2018

Report post

> **On 4/26/2018 at 2:46 PM, Manwith Noname said:**
>
> @bcink I have a question actually. How much work would be involved in copying a collider from one model to another?
>
> More generally, I'm curious if people look at their craft closely enough that differences like this would benefit them?

I'm not aware of the possibility of that type of a change.

Maybe on the differences. I lean towards probably.

Quote

Manwith Noname likes this

James Webb [WIP] Thread Kerbalow Aerospace [WIP] Thread DSSHU [WIP] Thread Making Alternate History [WIP] Thread Duna Direct [WIP] Thread

Manwith Noname

Rocketry Enthusiast

Posted April 26, 2018

Report post

> **On 4/26/2018 at 5:48 PM, Shadowmage said:**

Members
219
320 posts

> Not going to be doable without custom plugin code. Stock code doesn't support that level of model manipulation as far as I'm aware. You -might- be able to pull it off with one of the PartSwitch modules/mods (e.g. B9PartSwitch), but this would highly depend on exactly how the stock models are setup, and how the part-switch module handles sub-meshes.

Yeah. From what I can gather, the BW model contains the collider mesh, part variants keeps that but switches the render mesh. I was more thinking along the lines of exporting the BW model to blender, grabbing the collider mesh and placing it in to the O model, then porting that back in to the game. I've never dabbled with 3D modelling so I don't know how lengthy a process that would be or if it is even feasible. The MM patch I used works to make each model a seperate part but with no collider, you cannot interact with the orange tank model after it has been placed in the editor.

> **On 4/26/2018 at 5:48 PM, Shadowmage said:**
>
> After I'm finished updating SSTU for KSP 1.4+ I intend on digging into a couple of the larger outstanding issues with TU -- DirectX reflections, and the new stock part-variant mess (yes, it is a huge mess). So I would say hold off on the part-variant stuff for a bit longer, as I'll hopefully get some sort of texture switch (or just a recoloring) module developed for them in the (near'ish) future.

I was leaning towards not worrying about it too much for now, just curious if the workaround I imagined was easy short term (making a seperate model with a collider). For the base game, it's only a few parts that become a problem. I would certainly appreciate any dark magic waved in its direction but I wouldn't expect any rush to do so, particularly with all the projects you have going on.

> **On 4/26/2018 at 6:14 PM, bcink said:**
>
> Maybe on the differences. I lean towards probably.

Thanks. I'm just trying to gauge if people in general study the parts graphically when playing as much as things like this stand out when you're mucking around with things.

Quote

Money doesn't make the world go round...

...gravity does.

bcink

JWST Kerbal Engineer

Members

1,377

438 posts

Posted April 26, 2018

Report post

> **On 4/26/2018 at 6:59 PM, Manwith Noname said:**
>
> I was more thinking along the lines of exporting the BW model to blender, grabbing the collider mesh and placing it in to the O model, then porting that back in to the game.

If you were to create a new part that contains the collider required then that would be no problem, using the new part for the base of your variants.

Quote

James Webb [WIP] Thread Kerbalow Aerospace [WIP] Thread DSSHU [WIP] Thread Making Alternate History [WIP] Thread Duna Direct [WIP] Thread

Delbrutis

Curious But not George

Posted May 1, 2018

Report post

Members
44
128 posts

> On 4/26/2018 at 2:46 PM, Manwith Noname said:
>
> More generally, I'm curious if people look at their craft closely enough that differences like this would benefit them?

I would say on a part like that no one would notice. But that would totally be dependent on the part. And frankly I don't think messing with colliders would be a simple fix. Trust me I know the temptation Squads new texture switch looked sweet when I first saw it. But it's turned into a huge pain in my A$$. Plus messing with colliders on a part like a nose cone could end up having a huge impact on aero forces in the game. Squad broke a bunch of aero stuff with their update. And one would assume they should know what their doing? I think I saw someone report a bug where changing just the colour on a tank with the new switcher had a 7% change in drag. "That's some thick paint " I am really tired of messing with Squads new switchers. I'm going to apply TU to what base things I can. All the ComPods don't have switching "Cheep Window Shine" And the base metal stuff and FoilTanks & such. Then we will have to wait and see what happens? Good luck.

Quote

Manwith Noname

Rocketry Enthusiast

Members
219
320 posts

Posted May 4, 2018

Report post

@Delbrutis

I wasn't thinking about altering the nose cone collider. It's purely about stock part variants and how the models switch but it was just a thought for a quick fix and probably something I will not be looking at any further.

As for the textures, it has come about due to me noticing something that has been occurring for some time it seems when I looked further. The paint masks have always been in PNG format for a specific reason, accuracy. At some point post 1.0 KSP included a texture converter so now any texture that is not in DDS / DXT format gets converted at loading. This causes a problem...

Reveal hidden contents

The "blending" in low resolution textures converted to DXT / DDS is terrible. So, the solution is to make files in higher resolution as the effects become less noticeable. While the base textures don't need to be included in this, it kinda makes sense so the masks and maps all match. Also, it will help with some of the horrendous side effects of this appearing in other textures but that will require some handy work to remove as they are already baked in.

So, right now, I'm inclined to put together a "light" pack, which is native to stock and only a few megabytes in size. It will have these problems but will not take as much resources to run. Then as time goes on, a "heavy" pack will come together. It will be "HD" so will require more resources. Not exactly sure how much more yet but unless it is made to replace stock files, there will essentially be double the number of textures but the replacements will be approximately 16x the file size, maybe 4x in some cases. This of course doesn't include extra spec / met maps and paint masks...

Anyway, after some more dabbling with the nose cone...

Reveal hidden contents

Quote

Delbrutis likes this

Money doesn't make the world go round...

...gravity does.

Delbrutis

Curious But not George

Posted May 5, 2018 (edited)

Report post

The DDS / DXT problem has been a issue for awhile for people loading suits and custom textures with TRR. Squad went to the texture compression at 1.0 to help with the lower end marginal PC setup and speed up game loading DDS quality has been pretty horrible. And I think the consensus over on the TRR thread has been for

Members
44
128 posts

Texture&Suit authors to mostly just eat the extra overhead and use PNG's instead. So I can see what your doing sounds like the only solution that makes sense. I know you could never release a mod that does this but I'm curious. have you tried or do you know if the Squad textures can be replaced with a different texture format and still load properly ? Either way this is looking good so far and squads stock textures are pretty rough and pixelated anyway. So what people get is what they get. Wish I had something constructive to say but I don't Know? Good luck.

Edited May 5, 2018 by Delbrutis
Fix my crappy punctuation

Quote

Manwith Noname
Rocketry Enthusiast

Members
219
320 posts

Posted May 5, 2018 (edited)

Report post

> **On 5/5/2018 at 1:40 AM, Delbrutis said:**
>
> have you tried or do you know if the Squad textures can be replaced with a different texture format and still load properly ?

Up until about an hour ago no. However, I was scanning my log for errors when I stumbled on this...

```
[WRN 06:23:28.475] Texture resolution
is not valid for compression:
'E:\Software and
Installers\Games\KSP\Backups\Game\1.4\K
erbal Space
Program\GameData\Squad\Tutorials\YPRDia
gram.png' - consider changing the
image's width and height to enable
compression
```

DXT needs to be powers of 2 I believe. Not sure how much trimming a pixel of the right hand edge and bottom of the texture would cause a problem but it appears you can

"hack" around the compression by using non DXT compliant dimensions. Not tested it myself and probably will not for this project but it's certainly interesting.

Edited May 5, 2018 by Manwith Noname

Quote

Money doesn't make the world go round...

...gravity does.

Delbrutis

Curious But not George

Members

44

128 posts

Posted May 5, 2018

Report post

> **On 5/5/2018 at 3:53 AM, Manwith Noname said:**
>
> Up until about an hour ago no. However, I was scanning my log for errors when I stumbled on this...
>
> ```
> [WRN 06:23:28.475] Texture resolution is not valid for compression: 'E:\Software and Installers\Games\KSP\Backups\Game\1.4\Kerbal Space Program\GameData\Squad\Tutorials\YPRDiagram.png' - consider changing the image's width and height to enable compression
> ```
>
> DXT needs to be powers of 2 I believe. Not sure how much trimming a pixel of the right hand edge and bottom of the texture would cause a problem but it appears you can "hack" around the compression by using non DXT compliant dimensions. Not tested it myself and probably will not for this project but it's certainly interesting.

That's Interesting thanks.

Quote

Rallyman03
Bottle Rocketeer

Members
1
16 posts

Posted May 9, 2018

Report post

> **On 4/24/2018 at 6:46 PM, Manwith Noname said:**
>
> Essentially, yes, though that is a part of the "thinking in progress" title. I was going to add a roadmap to the opening post at some point when I've though it out some more but off the top of my head it looks something like this...
>
> Phase 1 - Get maps made / tidy for parts and working in game.
>
> Phase 2 - Create dedicated specular / metallic maps.
>
> Phase 3 - Normals?
>
> There's also something I need to decide before starting phase 2 in earnest. Textures for masked shading without "tinting mode" appear to all be scaled around 127 / 128 RGB values while the tinting mode expects the max value to be 255...though I might be mistaken and misread things.
>
> After converting configs to work in 1.4 I started dabbling in Phase 2 but I will probably revert to how things were working prior for any test release as converting every parts textures is quite a task to do and while without them there are issues, having a mismatch of how parts and textures are setup is probably more annoying. I should perhaps also mention, while getting stock sorted should probably be the priority, I have some mods covered too and would like to cover more.
>
> **Reveal hidden contents**

You have no idea how excited I am for the OPT Space Plane shaders. I build pretty much exclusively with those parts at this point.

Quote

Manwith Noname likes this

Cpt_MarcosSilva

Bottle Rocketeer

Members

1

11 posts

Posted May 10, 2018

Report post

I'm working in a very similar project, but making new PSD layers from every single texture is very time consuming, hope you finish your project.

Congratulations.

Quote

Manwith Noname likes this

Manwith Noname

Rocketry Enthusiast

Members

219

320 posts

Posted May 11, 2018 (edited)

Report post

> **On 5/9/2018 at 7:04 PM, Rallyman03 said:**
>
> You have no idea how excited I am for the OPT Space Plane shaders. I build pretty much exclusively with those parts at this point.

I've got a little sidetracked by something so I've not done anything with regard to updating the OPT parts but a little heads up, it's only the base OPT. I did start looking at the legacy pack and covered a couple but by the time I got to them I was little bit burned out. This was back in January / February. I'll look at getting an OPT pack set up when I've got a test release out for stock...which should hopefully be soon.

> **On 5/10/2018 at 4:44 AM, Cpt_MarcosSilva said:**
>
> I'm working in a very similar project, but making new PSD layers from every single texture is very time consuming, hope you finish your project.
>
> Congratulations.

Nice. Are you working on mods or stock? I never got round to putting it in the opening post but part of my thinking for this thread was to encourage people to come forward and mention projects they might be working on. Would be better if people spread the work load out in a way. I've also been thinking about trying to drive some sort of community based stock texture switch standardisation because if everyone is left to their own devices, we're going to end up with a load of packs that can't be used together. I say this mainly because I've done some things where possible to allow more than 3 colours on a part and it will either break other patches or other patches will break it if things are not set up correctly. I'd also been toying with the idea of making multiple designs for stock but...I haven't finished the first set yet.

Edited May 11, 2018 by Manwith Noname

+ Quote

Money doesn't make the world go round...

...gravity does.

Cpt_MarcosSilva

Bottle Rocketeer

Members

1

11 posts

Posted May 12, 2018

Report post

On 5/11/2018 at 4:26 PM, Manwith Noname said:

Nice. Are you working on mods or stock? I never got round to putting it in the opening post but part of my thinking for this thread was to encourage people to come forward and mention projects they might be working on. Would be better if people spread the work load out in a way. I've also been thinking about trying to drive some sort of community based stock texture switch standardisation because if everyone is left to their own devices, we're going to end up with a load of packs that can't be used together. I say this mainly because I've done some things where possible to allow more than 3 colours on a part and it will either break other patches or other patches will break it if things are not set up correctly. I'd also been toying

> with the idea of making multiple designs for stock but...I haven't finished the first set yet.

The purpose of the project is actually make new textures and maps using the stock textures as a base to make it locks realistic with the metallic shader, making new ambient occlusion maps, normal maps and of course, the MET mask. I have some success with the MK3 fuselage, the fuselage plates looks that are really separate parts that was mounted together, not a single part with painted plates as the stock part is.
Unfortunately I delete everything as I know that I could make better quality maps using both Illustrator and photoshop to do that, maybe by the week I post some screenshots

+ Quote

Electrocutor
Tinkerer
●●●●

Members
600
617 posts

Posted May 17, 2018

Report post

It's always difficult to maintain motivation on such a huge project; might I suggest breaking it down into sub-releases until everything is eventually done? For example, you seem to have the majority of mk2 parts completed or mostly complete; so why not release mk2 only to start, then get feedback and work your way through iterations until you are pleased with the final product of mk2; then proceed to give the same treatment to other groups?

As for compression issues and performance issues using high resolution textures... there is really only one true solution to that problem and that is to make the models for parts the way they should have been to start with: each material/color separated into their own sub-mesh and use a single set of small tiled/seamless textures shared across all parts. This is, unfortunately, a huge model undertaking; so for now at least, just use what you have. Most people are already used to KSP's lack-luster graphics, so simply having a full implementation of recoloring options that is easy to install and use is their biggest desire.

As a side note, you may want to get in touch with @DoctorDavinci to combine recoloring efforts (from DCK). I expect he wouldn't mind having metallic and reflective options alongside his existing recolors, and TU recolor has support for patterning. This may also help you to figure out how to setup MM patches that allows others to add additional color or pattern presets without interfering with existing ones. My guess is that you would want to integrate with the PartVariants vanilla module as a common-ground, then have the recolor GUI available for customization.

Keep in mind that you can set any material properties within a PartVariants TEXTURE node, such as...

```
VARIANT {
        TEXTURE {
                _MainTex =
                _MaskTex =
                _MaskMetallic =
                _MaskColor1 =
                _MaskColor2 =
                _MaskColor3 =
        }
}
```

Quote

Jesusthebird

Lander Commander

Posted July 3, 2018

Report post

Is the color mapping part of this project or a separate plugin perhaps? I'm looking for some sort of teaching outline for parts; similar to TRR_Guide. This reminds me of that mod in that sense, and was hoping I could use it as a learning reference.

Members
126
501 posts
Location: LKO

your stock variant looks great by the way. I wish you much luck in this project.

Quote

alexus
Spacecraft Engineer

Members
51
167 posts

Posted September 6, 2018

Report post

Can you get us config sample?

Quote

It`s my mods:

- Milky way skycube 402 Mpx!
- Textures Unlimited Configs and Retexturing (TUCaR)

Manwith Noname
Rocketry Enthusiast

Members
219
320 posts

Posted September 23, 2018 (edited)

Report post

Alright, not going to go in to all the details but after massive amounts of procrastination, some downloads are up. Some people may be disappointed in the not so massive leap from what was available in 1.3 but this is mainly a "inb4 KSP1.5" breaks it...potentially massively. So it might aswell get some use.

There are certainly some new parts covered, some tweaks to previously covered parts....some mod packs...some more in the pipeline...maybe.

Edit:...oh, and some more editing of the OP, at some point. Gotta make some eye candy.

Another edit: ....just remembered something I meant to do before packing them up...oh well. I know of at least one thing that isn't right.

Edited September 23, 2018 by Manwith Noname

Quote

Delbrutis likes this

Money doesn't make the world go round...

...gravity does.

Electrocutor

Tinkerer

Members

600

617 posts

Posted September 23, 2018

Report post

Very nice. Thank you.

Quote

Jimbodiah

SSTU Fanboy

Members

1,385

2,325 posts

Posted September 23, 2018

Report post

Awesome!!!!

Quote

JPL Jimbodiah's Patch Laboratory

RSS Simple patch

SSTU Color Presets

1 2 3 4 5 6 NEXT Page 1 of 7

Reply to this topic...

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